BRAC UNIVERSITY

Inspiring Excellence

CSE320 ASSIGNMENT 01

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Section- 13

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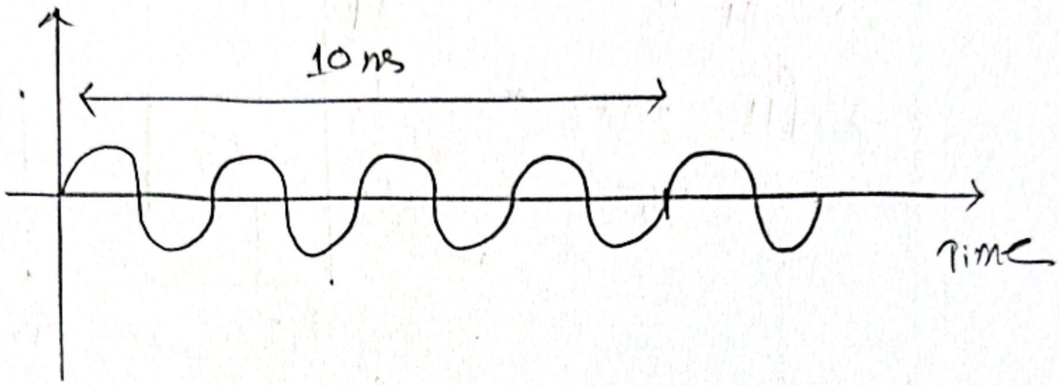
Ans. to the Quest. no-01



$10\times10^{-4}s$ makes 4 cycles

$1s$ makes $\frac{4}{10\times10^{-4}}$ cycles

$= 4\times10^{8}$ cycles

$\therefore$ frequency $= 4\times10^{8}$ Hz

and time period $= \frac{1}{f}$

$= \frac{1}{4\times10^{8}} = 2.5\times10^{-9}s$

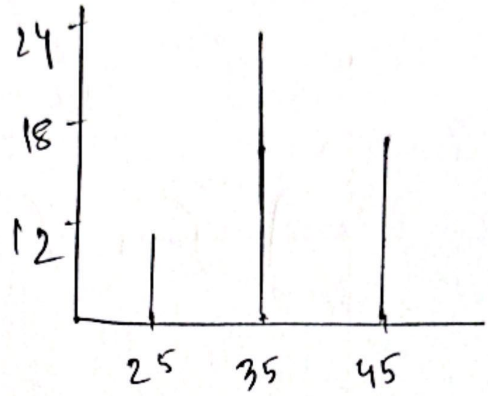
In the 2nd diagram, we can see a phase shift of the wave. This sort of transmission impairment is called distortion.

Ans. to the Quest no-02

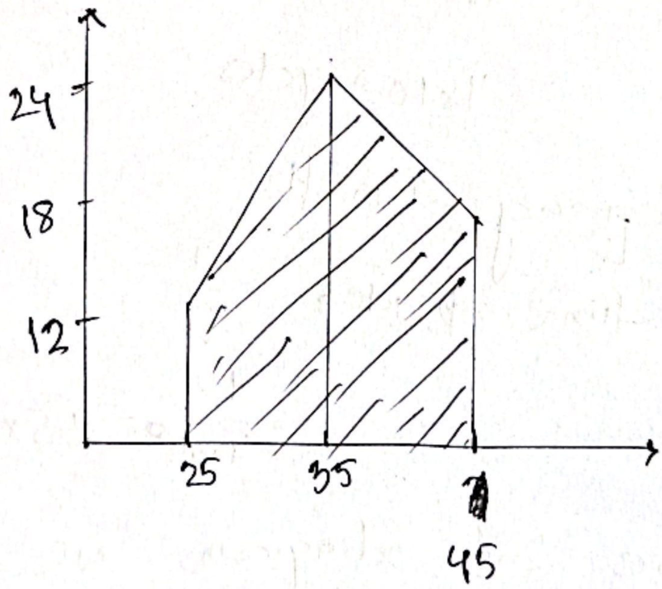
20 = f_H - 25

$\Rightarrow f_H = 20+25$

$= 45$



Ans. to the Quest. no-03



Ans. to the Quest. no-04

(1) $20 \times 1080 \times 24 \times 60 \times 0.05$

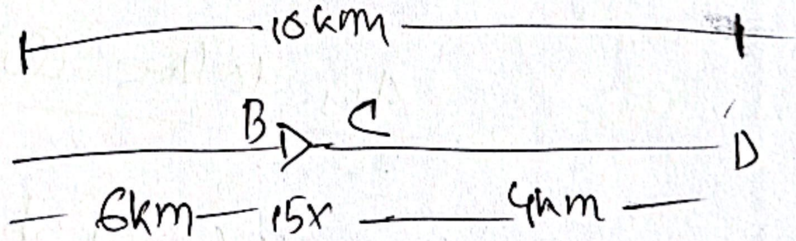
$= 149299200 bps$

$= 149299.2 kbps$

$= 149.2992 \ mbps$

(Ans)

Ans. to the Quest. no-05



$3W/km$

$\therefore P_B = 20 - (3 \times 6) = 20 - 18$

$= 2W$

$\therefore$ atten $AB = 10 \log_{10} \frac{P_B}{P_A} = 10 \log_{10} \frac{2}{20} = -10$

$P_C = 2 \times 15W = 30W$

$\therefore$ atten $BC = 10 \log_{10} \frac{P_C}{P_B} = 10 \log \frac{30}{2} = 11.76$

So. at D point $D = 30-(3\times4)$

$= 18W$

$\therefore$ atten CD $= 10\log_{10}\frac{PA}{PC}$

$= 10\log_{10}\frac{18}{30}$

$= -2.218$

$\therefore$ total attenuation $= -10+11.76-2.218$

$= -0.458$ (Ans)

Ans. to the Quest. no 6

data rate $= 75\%$ of shanon

$= 0.75C$

$\therefore$ bitrate $= \frac{1.35\times10^8\times8}{5\times60} = 3600000$

$\therefore C = \frac{\text{bitrate}}{0.75} = \frac{3600000}{0.75}$

$= 4.8\times10^6$ bps

$= 4.8$ mbps

Now,

$$C = B\log_2(1+SNR)$$

$$\Rightarrow SNR = 2^{(C/B)} - 1$$

$$= 2^{\left(\frac{4.8\times10^6}{100\times10^3}\right)} - 1$$

$$= 2.814\times10^{14}$$

and, $2B\log_2 L = 3.6\times10^6$

$$\therefore L = 2^{\left(\frac{3.6\times10^6}{2\times100\times10^3}\right)} = 262144$$

(Ans)

Ans. to the Quest. no-07

$$\text{Queue} = (10\times3\times10^{-6}) = 3\times10^{-5}$$

$$\text{Propagation delay} = \frac{2000\times10^3}{2\times10^8}$$

$$= 0.01s$$

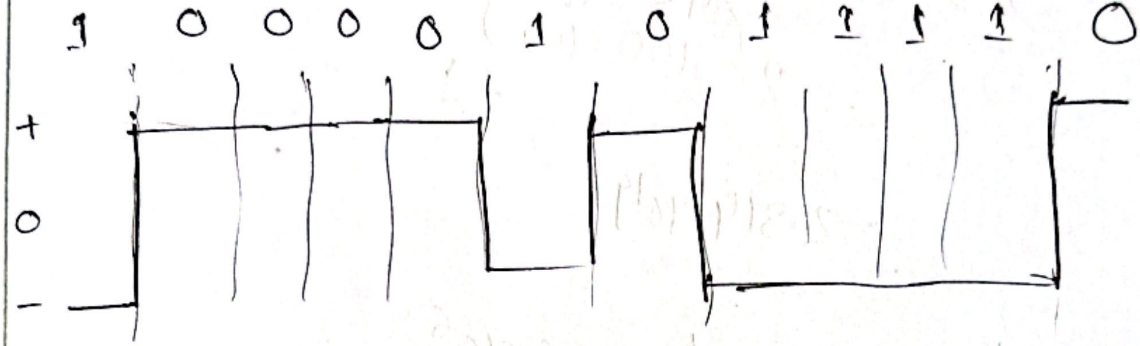
$$\text{Transmission delay} = \frac{5\times10^6\times8}{5\times10^6} = 8s$$

$$\therefore \text{Total latency} = 8+0.01+3\times10^{-5}$$

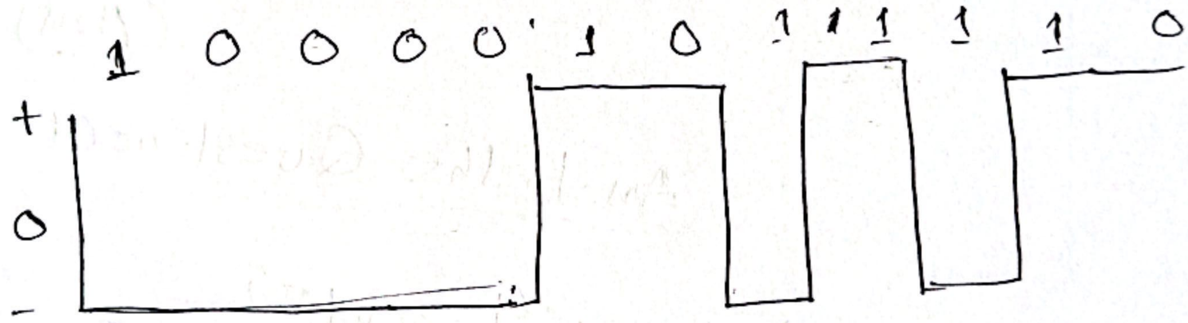
$$= 8.01003s$$

Ans. to the Quest. no-08

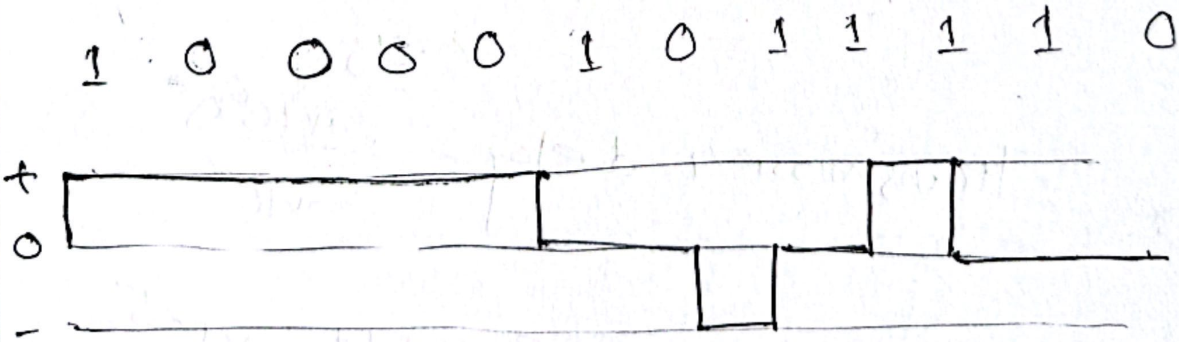
NRZ-L:



NRZ-I:

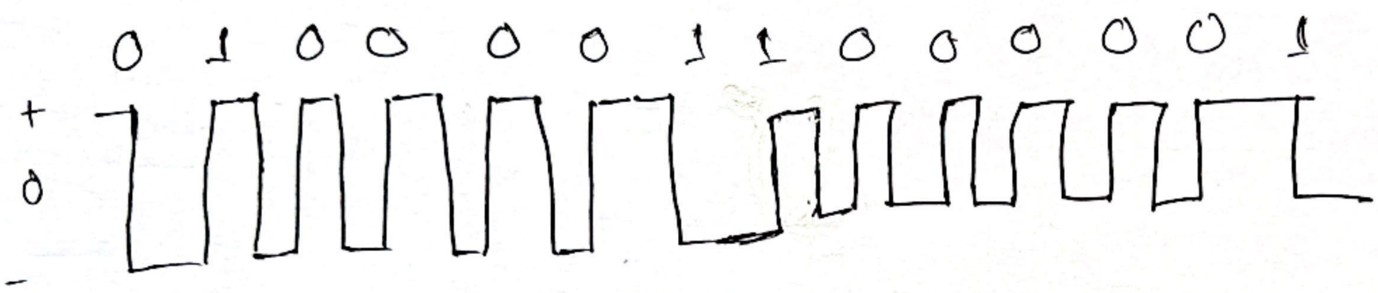


MLT-3:



(AM)

Now, we will use DMAN

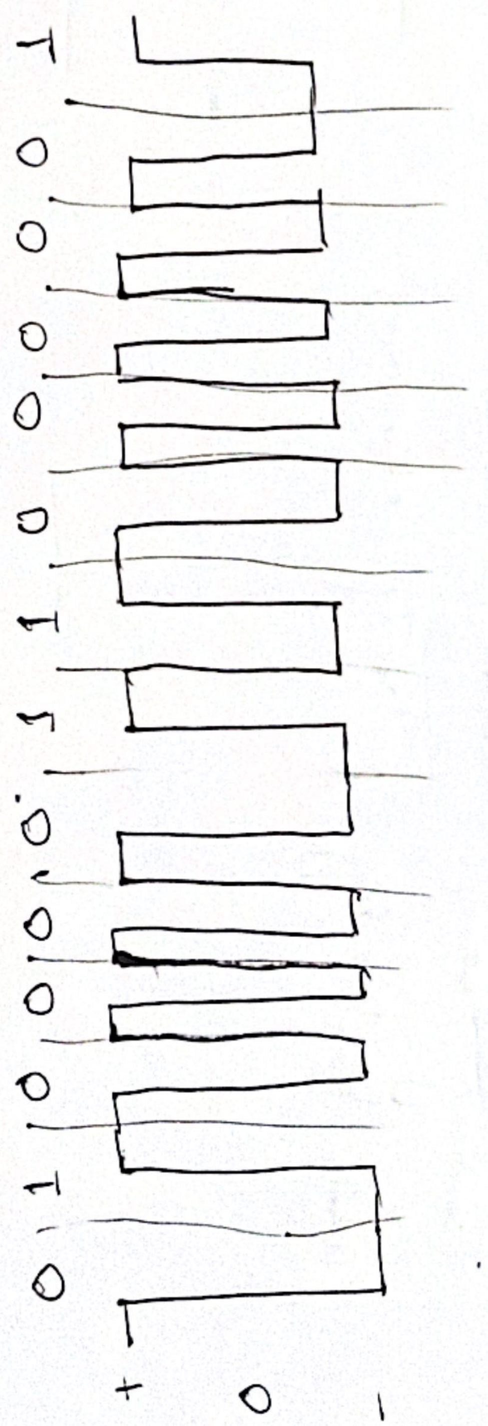


Among those DMAN is better as it's not affected by polarcity changes.

Ans. to the Question no-09

The code is: 0100001100001

We will use MAN



Ans. to the Quest. no- 10

Here,

MLT-3 was used.

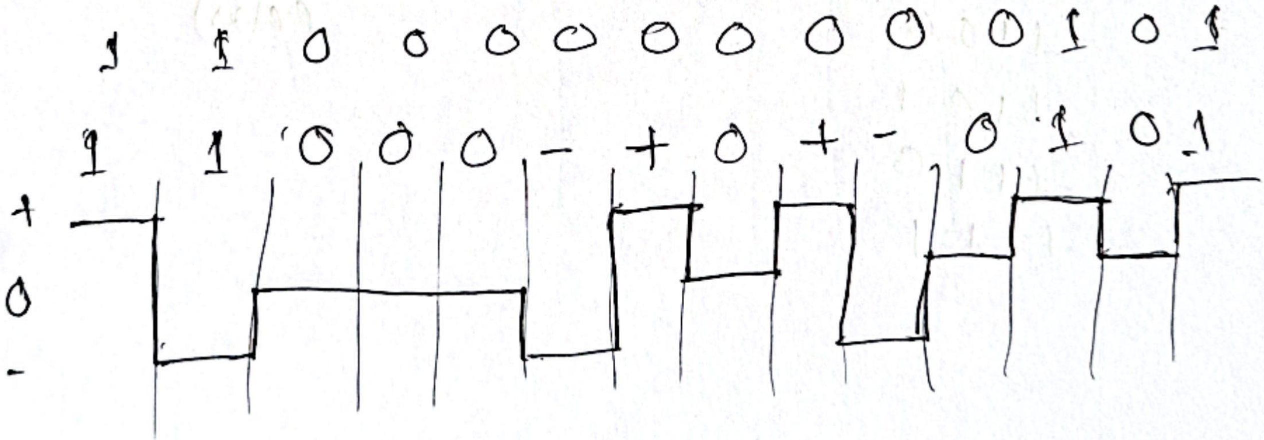
The signal was: 1 1 0 0 0 0 0 0 0 0 0 1 0 1

Issues are →

(i) Baseline wandering

(ii) DC component

(iii) lack of synchronization.



Ans. to the Quest. no-11

~~0000~~
~~0001~~
0010
0011
0100
0101
0111
~~1000~~
1001
1011
~~1100~~
~~1101~~
~~1110~~
~~1111~~

3B	4B
000	0010
001	0011
010	0100
011	0101
100	0111
101	1001
110	1010
111	1011

(Ans)